Week 8 – Storing Big Data

02613 Python and High-Performance Computing

“My job was killed because it used too much memory”
- Some of you

“Just request more memory”
- Me

“My job was killed because it used too much memory”

- Stud

What if you can’t?

ry”

- Me

Today

Handling data which does not fit into RAM

- Pandas chunking
- NumPy memory mapping
- Zarr arrays

Pandas chunking

Pandas chunking

```
>>> import pandas as pd
>>> df = pd.read_csv(‘bighuge.csv’)
>>> df[‘coordsx’].sum()
2661914.503899999
```

What if we cannot load the DataFrame?

Pandas chunking

```
>>> import pandas as pd
>>> dfc = pd.read_csv(‘bighuge.csv’, chunksize=100)  # Load 100 rows at a time
>>> dfc
<pandas.io.parsers.readers.TextFileReader object at 0x7fd5202f7790>
>>> for df in dfc:
...     print(df.shape, df.columns)
...
<Lots of output>
(100, 7) Index(['coordsx', 'coordsy', 'created', 'observed', 'parameterId',
'stationId', 'value'],
      dtype='object')
(55, 7) Index(['coordsx', 'coordsy', 'created', 'observed', 'parameterId',
'stationId', 'value'],
      dtype='object')
```

Pandas chunking

```
>>> import pandas as pd
>>> dfc = pd.read_csv('bighuge.csv', chunksize=100)  # Load 100 rows at a time
>>> dfc
<pandas.io.parsers.readers.TextFileReader object at 0x7fd5202f7790>
>>> coordsum = 0.0
>>> for df in dfc:
...     coordsum += df['coordsx'].sum()
...
>>> coordsum
2661914.5039
```

```
>>> df['coordsx'].sum()
2661914.503899999
```

Pro: don't have to hold full DataFrame in memory.

Con: more complex code

Pandas chunking

```
>>> import pandas as pd
>>> dfc = pd.read_csv(‘bighuge.csv’, chunksize=100)  # Load 100 rows at a time
>>> dfc
<pandas.io.parsers.readers.TextFileReader object at 0x7fd5202f7790>
>>> coordsum = 0.0
>>> for df in dfc:
...     coordsum += df[‘coordsx’].sum()
...
>>> coordsum
2661914.5039
>>> coordsum2 = 0.0
>>> for df in dfc:
...     coordsum2 += df[‘coordsx’].sum()
...
>>> coordsum2
0.0
```

Can only run through chunks once. Must then re-open reader.

Pandas chunking

```
>>> import pandas as pd
>>> dfc = pd.read_csv(‘bighuge.csv’, chunksize=100)  # Load 100 rows at a time
>>> dfc
<pandas.io.parsers.readers.TextFileReader object at 0x7fd5202f7790>
>>> dfc[1]
Traceback (most recent call last):
  File "<stdin>", line 1, in <module>
TypeError: 'TextFileReader' object is not subscriptable
>>> len(dfc)
Traceback (most recent call last):
  File "<stdin>", line 1, in <module>
TypeError: object of type 'TextFileReader' has no len()
```

No way to know how long original data is

Pandas chunking

```
1,2020-01-01 00:28:15,2020-01-01 00:33:03,1,1.20,1,N,238,239,1,6,3,0.5,1.47,0,0.3,11.27,2.5
1,2020-01-01 00:35:39,2020-01-01 00:43:04,1,1.20,1,N,239,238,1,7,3,0.5,1.5,0,0.3,12.3,2.5
1,2020-01-01 00:47:41,2020-01-01 00:53:52,1,.60,1,N,238,238,1,6,3,0.5,1,0,0.3,10.8,2.5
1,2020-01-01 00:55:23,2020-01-01 01:00:14,1,.80,1,N,238,151,1,5.5,0.5,0.5,1.36,0,0.3,8.16,0
2,2020-01-01 00:01:58,2020-01-01 00:04:16,1,.00,1,N,193,193,2,3.5,0.5,0.5,0,0,0.3,4.8,0
2,2020-01-01 00:09:44,2020-01-01 00:10:37,1,.03,1,N,7,193,2,2.5,0.5,0.5,0,0,0.3,3.8,0
2,2020-01-01 00:39:25,2020-01-01 00:39:29,1,.00,1,N,193,193,1,2.5,0.5,0.5,0.01,0,0.3,3.81,0
2,2019-12-18 15:27:49,2019-12-18 15:28:59,1,.00,5,N,193,193,1,0.01,0,0,0,0,0.3,2.81,2.5
2,2019-12-18 15:30:35,2019-12-18 15:31:35,4,.00,1,N,193,193,1,2.5,0.5,0.5,0,0,0.3,6.3,2.5
1,2020-01-01 00:29:01,2020-01-01 00:40:28,2,.70,1,N,246,48,1,8,3,0.5,2.35,0,0.3,14.15,2.5
1,2020-01-01 00:55:11,2020-01-01 01:12:03,2,2.40,1,N,246,79,1,12,3,0.5,1.7
1,2020-01-01 00:37:15,2020-01-01 00:51:41,1,.80,1,N,163,161,2,9.5,3,0.5,0,
1,2020-01-01 00:56:27,2020-01-01 01:21:44,1,3.30,1,N,161,144,1,17,3,0.5,4.
2,2020-01-01 00:21:54,2020-01-01 00:27:31,1,1.07,1,N,43,239,1,6,0.5,0.5,1.
2,2020-01-01 00:38:01,2020-01-01 01:15:21,1,7.76,1,N,143,25,1,28.5,0.5,0.5
1,2020-01-01 00:15:35,2020-01-01 00:27:06,3,1.60,1,N,211,234,2,9,3,0.5,0,0
1,2020-01-01 00:41:20,2020-01-01 00:44:22,1,.50,1,Y,234,90,1,4,3,0.5,1,0,0
1,2020-01-01 00:56:38,2020-01-01 01:13:34,1,1.70,1,N,246,142,2,11.5,3,0.5,0,0,0.3,15.3,2.5
2,2020-01-01 00:08:21,2020-01-01 00:25:29,1,8.45,1,N,138,216,2,24.5,0.5,0.5,0,0,0.3,25.8,0
1,2020-01-01 00:25:39,2020-01-01 00:27:05,1,.00,1,N,170,162,4,3,3,0.5,0,0,0.3,6.8,2.5
```

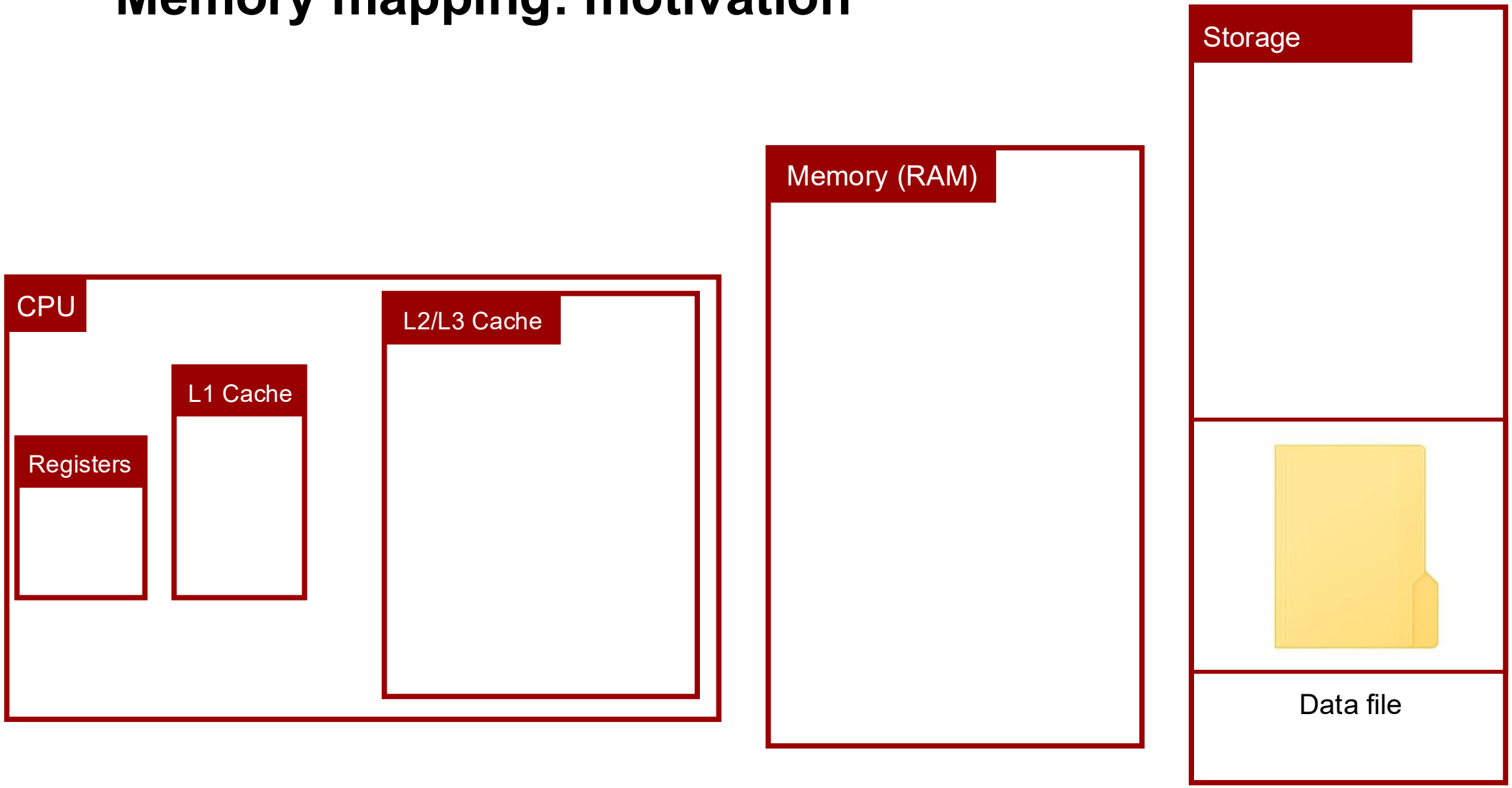
No way to know how long original data is, because each line is different

Parquet chunking

```
>>> import pyarrow.parquet as pq
>>> pf = pq.ParquetFile("bighuge.parquet")
>>> pf.num_row_groups
82
>>> g = pf.read_row_group(2)  # Read 3rd row group
>>> g.shape
(100, 7)
>>> df = g.to_pandas()  # Back to pandas
```

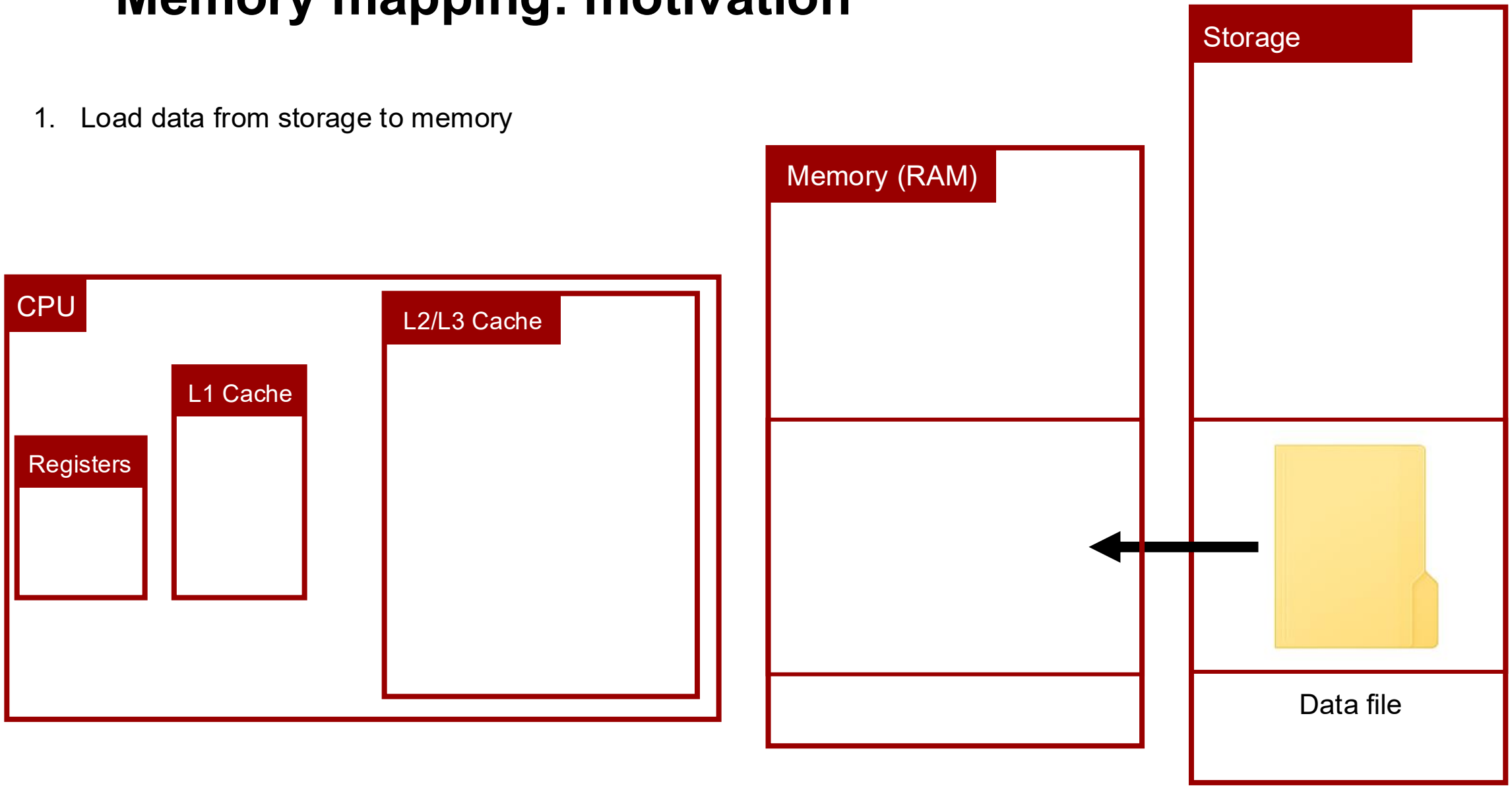
Memory mapping

Memory mapping: motivation



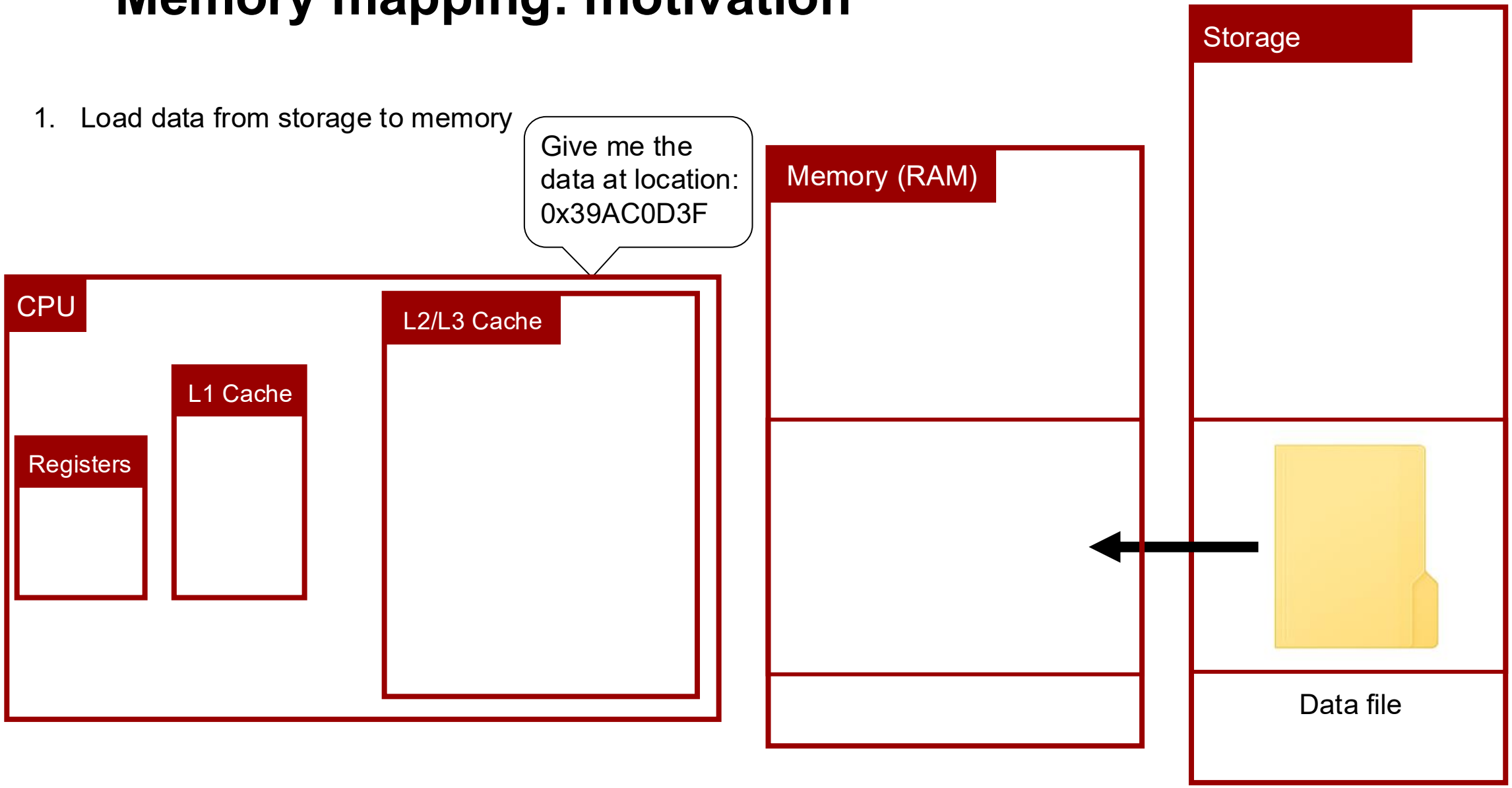
Memory mapping: motivation

1. Load data from storage to memory



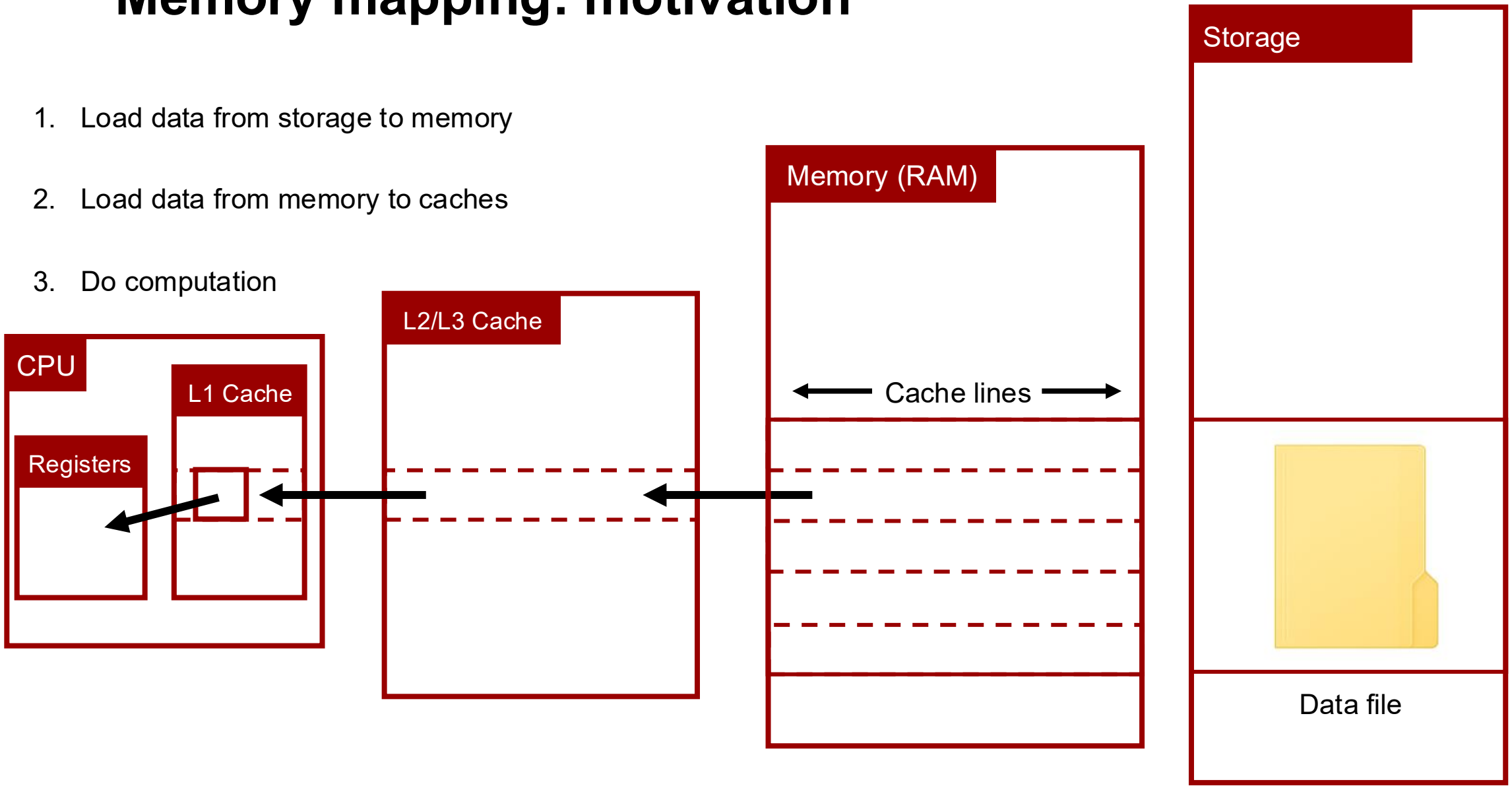
Memory mapping: motivation

1. Load data from storage to memory



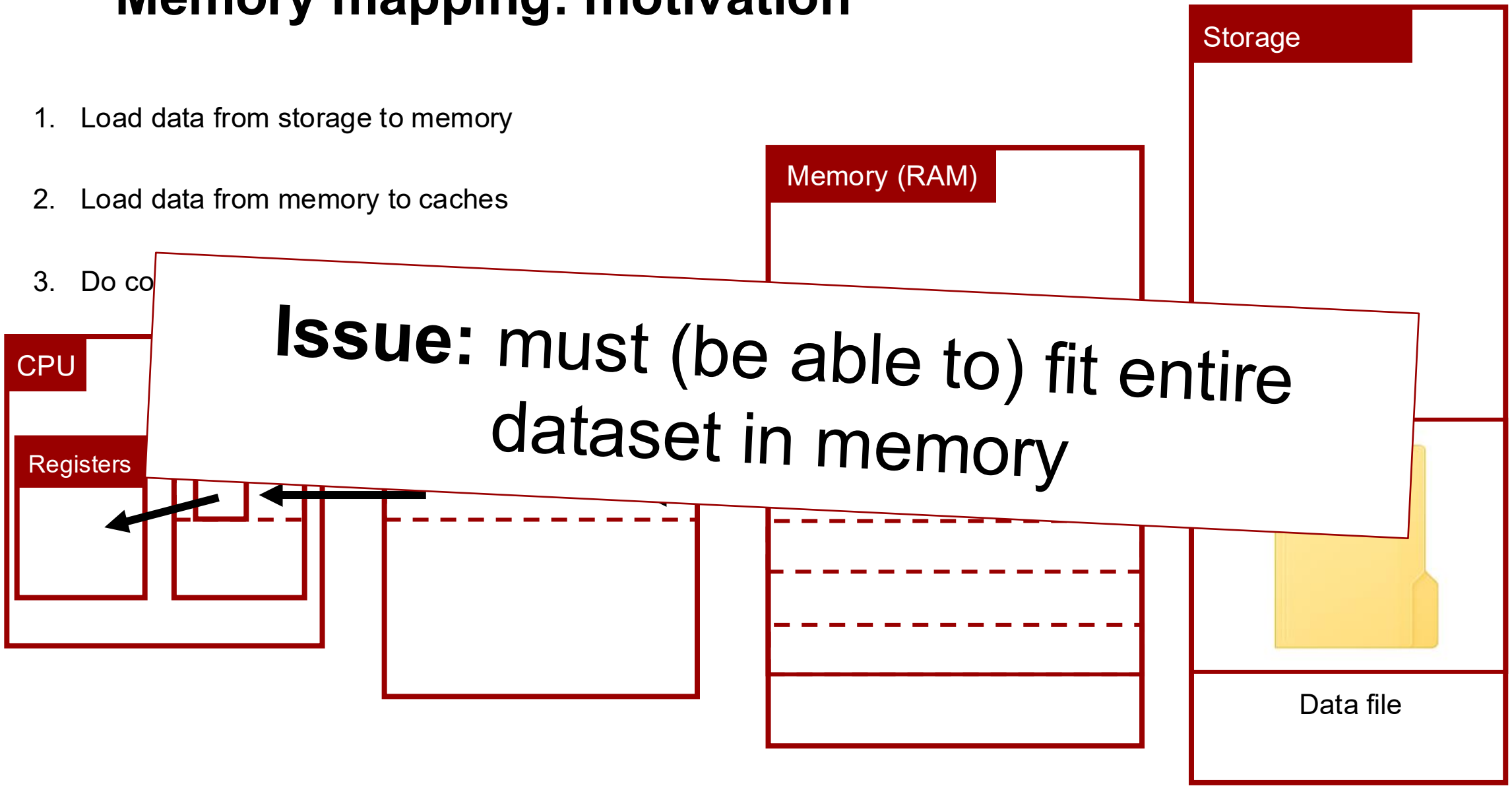
Memory mapping: motivation

1. Load data from storage to memory
2. Load data from memory to caches
3. Do computation



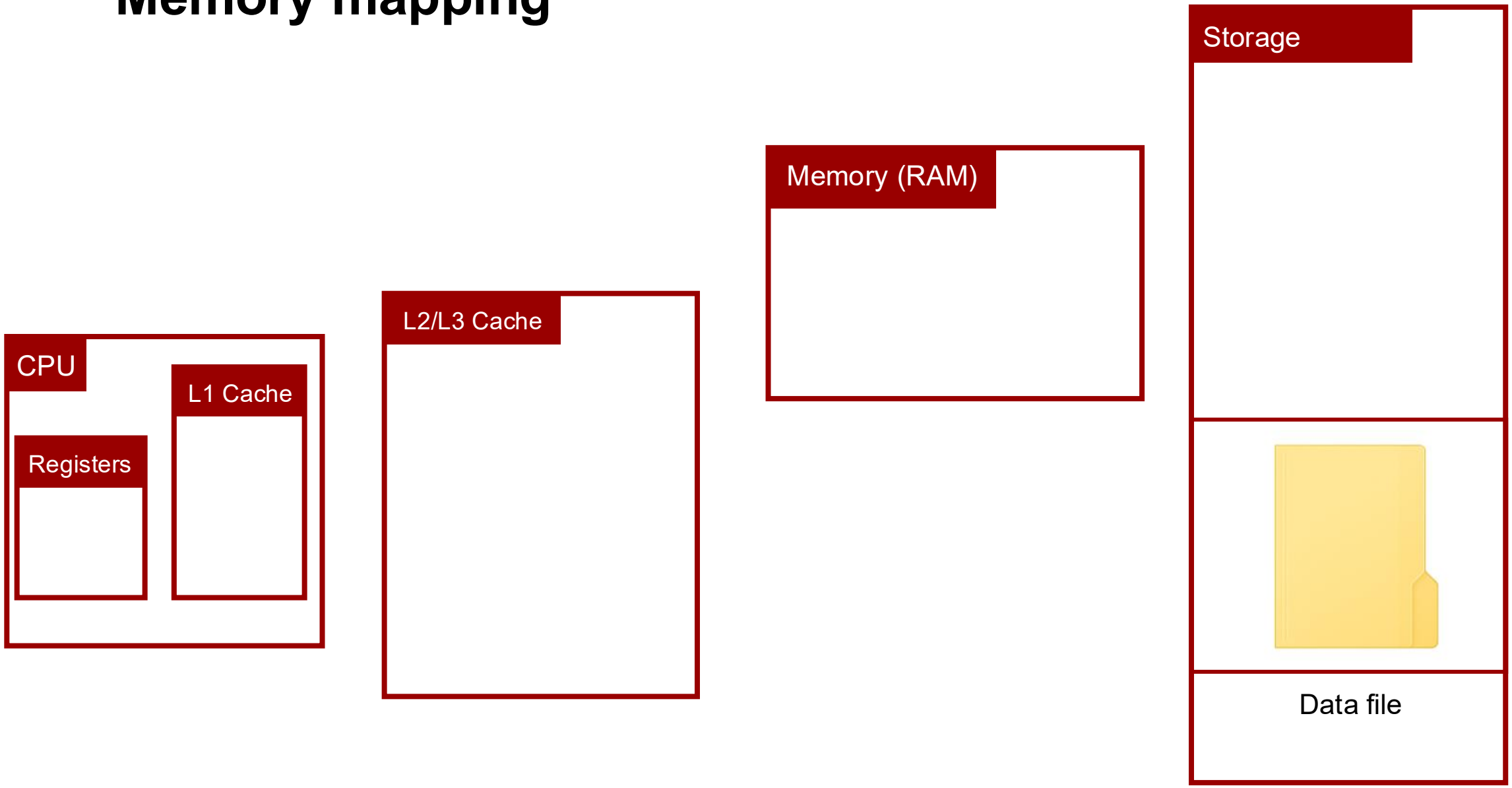
Memory mapping: motivation

1. Load data from storage to memory
2. Load data from memory to caches
3. Do co

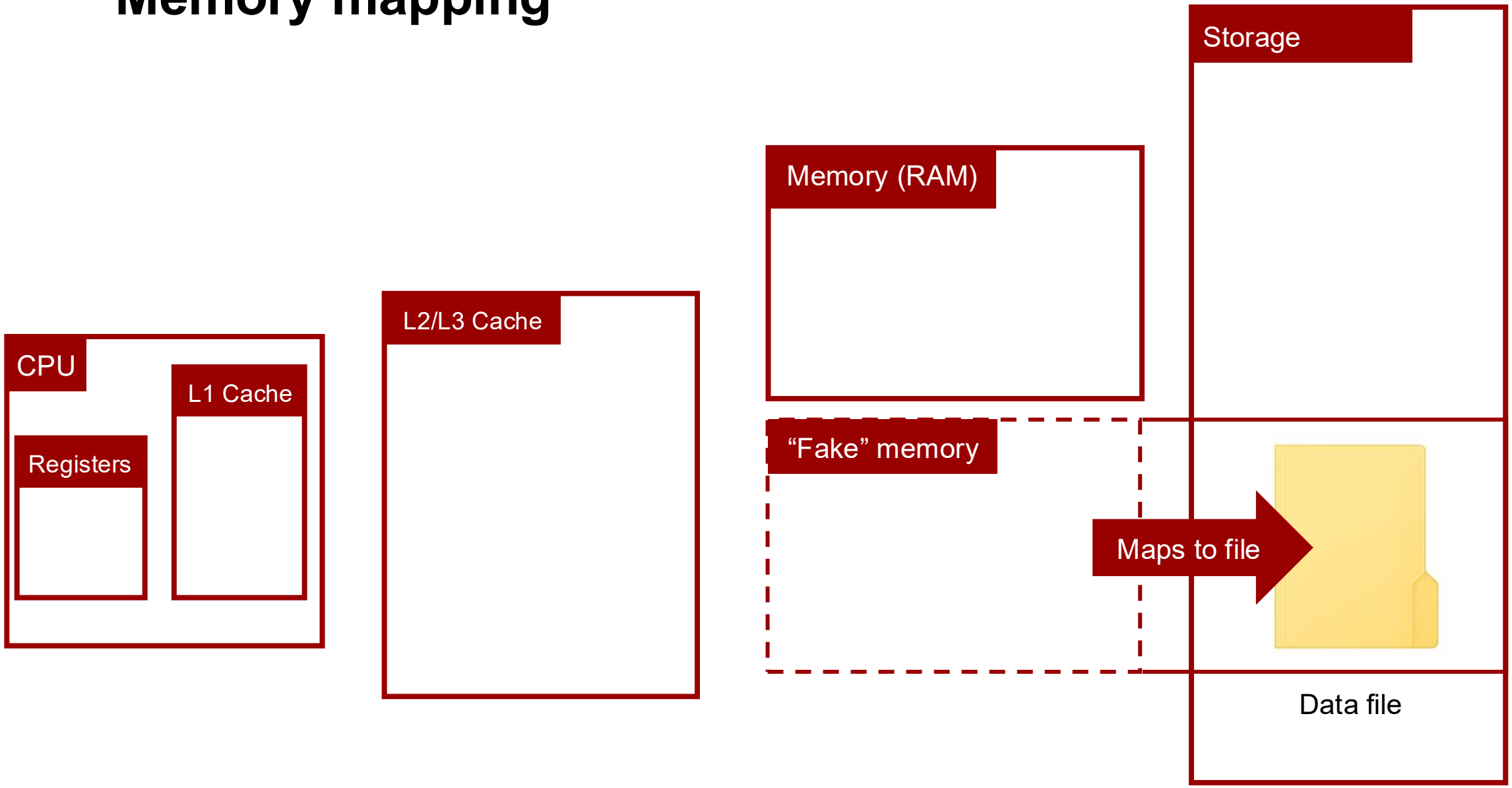


Issue: must (be able to) fit entire dataset in memory

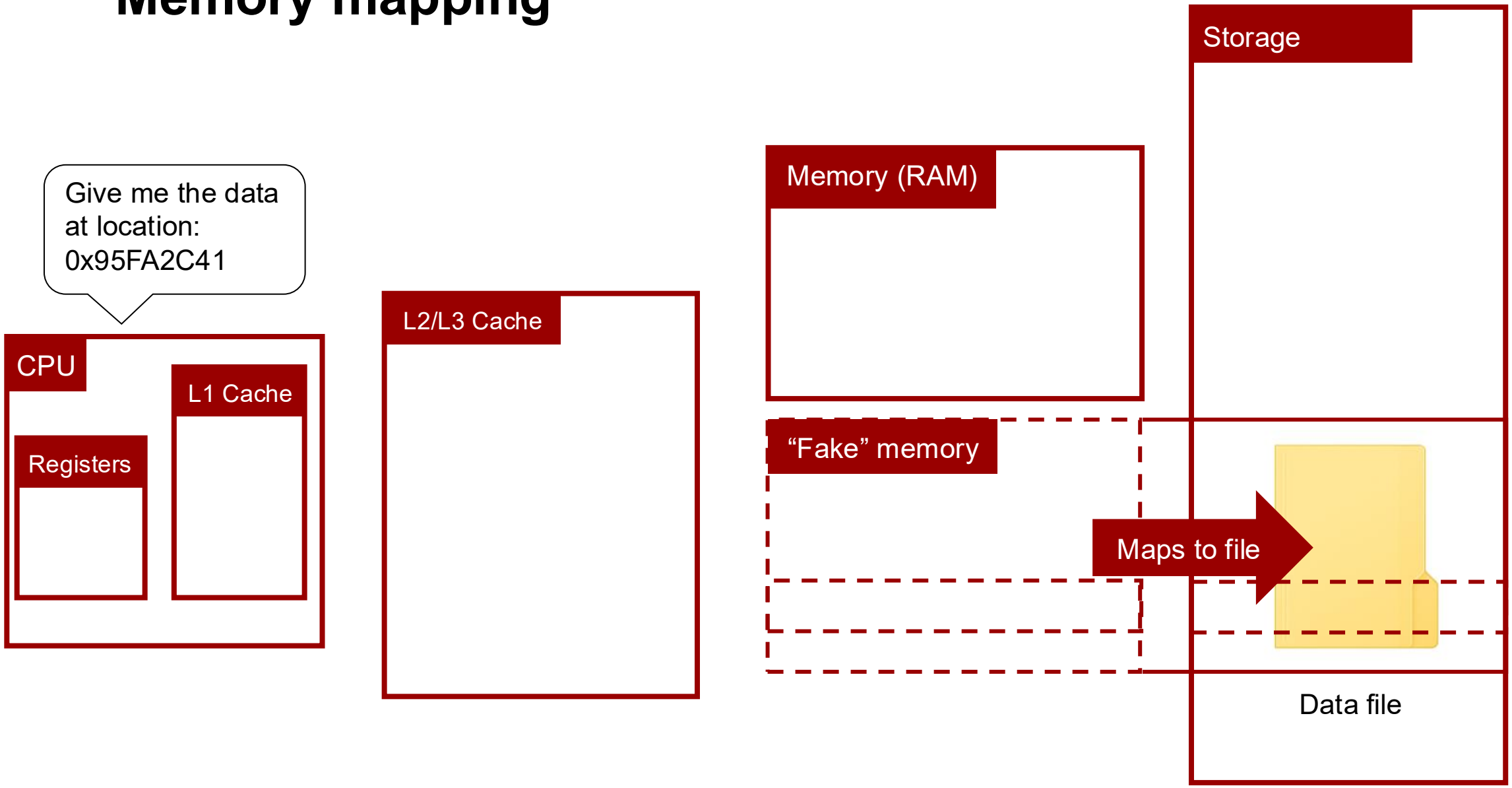
Memory mapping



Memory mapping



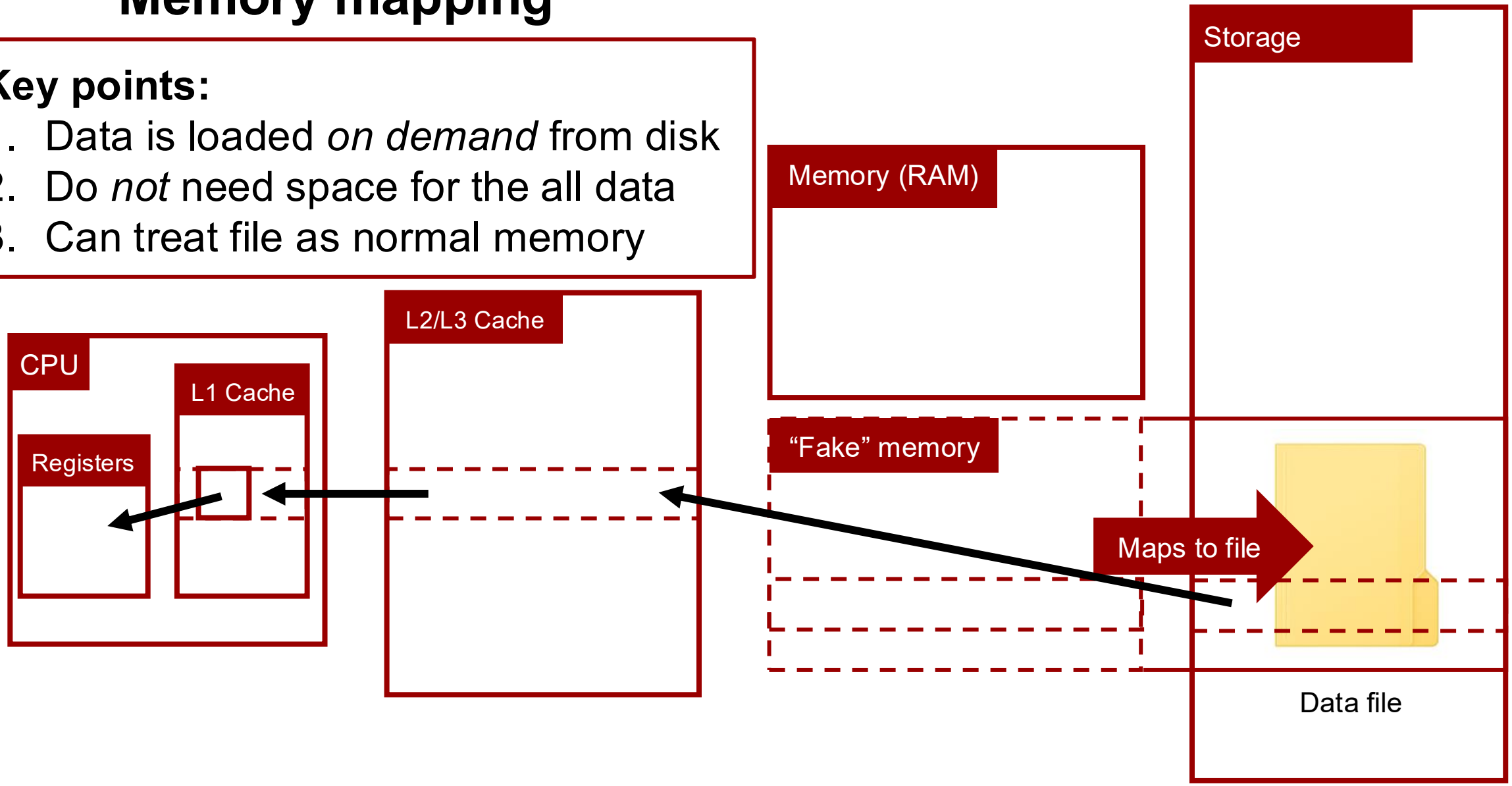
Memory mapping



Memory mapping

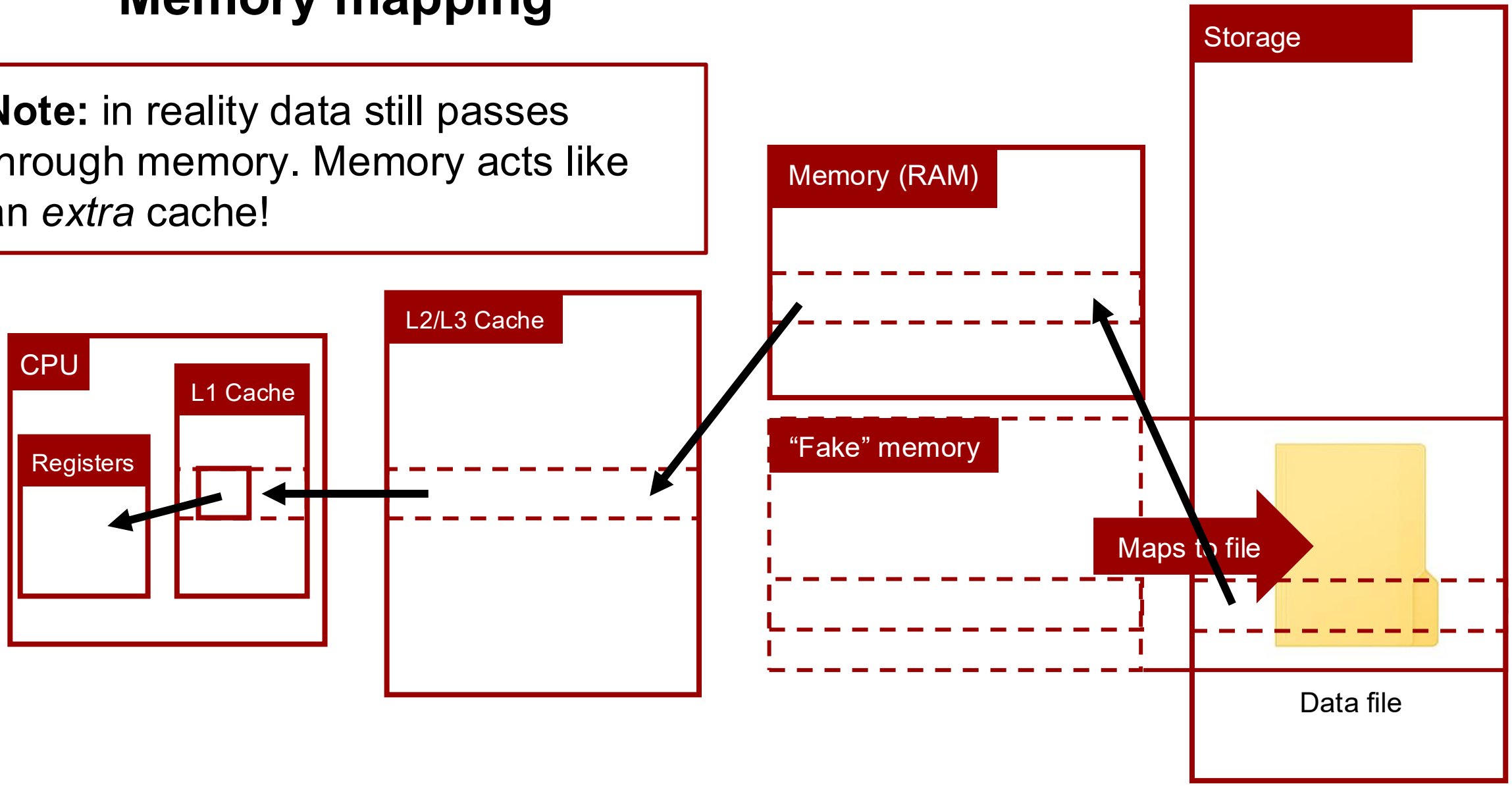
Key points:

1. Data is loaded *on demand* from disk
2. Do *not* need space for the all data
3. Can treat file as normal memory



Memory mapping

Note: in reality data still passes through memory. Memory acts like an *extra* cache!

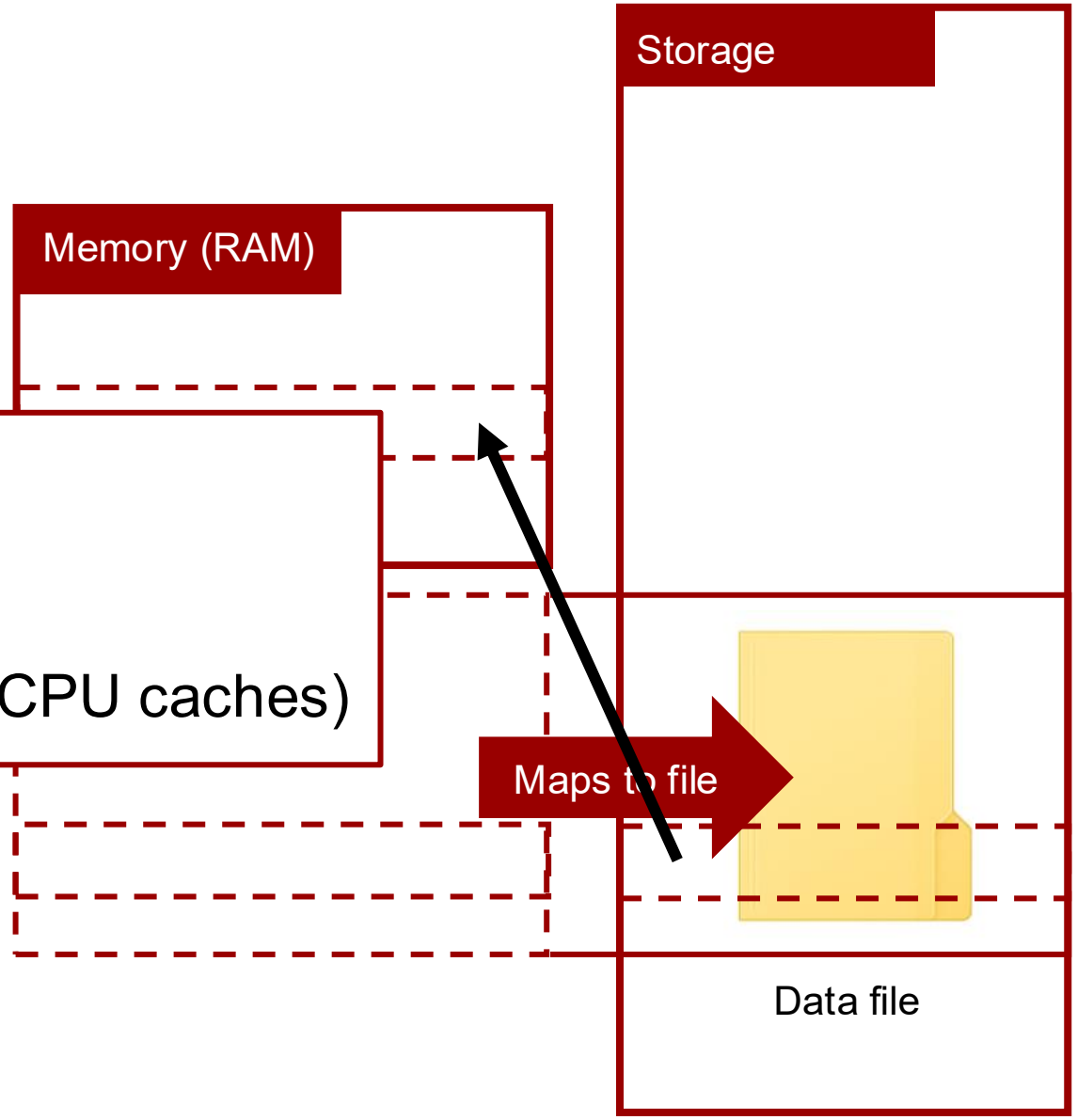


Memory mapping

Note: in reality data still passes through memory. Memory acts like an *extra* cache!

Consequences:

1. Reading / writing cached data = fast
2. Reading / writing from file = slow
3. Data evicted once memory is full (like CPU caches)



Memory mapping: NumPy

```
>>> import numpy as np
>>> mm = np.memmap('tmp.raw', mode='w+', shape=(10, 10), dtype='float64')
```

File name | Mode | Array shape | Data type

‘w+’ = create / overwrite file

Memory mapping: NumPy

```
>>> import numpy as np
>>> mm = np.memmap('tmp.raw', mode='w+', shape=(10, 10), dtype='float64')
>>> mm
memmap([[0., 0., 0., 0., 0., 0., 0., 0., 0., 0.],
        [0., 0., 0., 0., 0., 0., 0., 0., 0., 0.],
        [0., 0., 0., 0., 0., 0., 0., 0., 0., 0.],
        [0., 0., 0., 0., 0., 0., 0., 0., 0., 0.],
        [0., 0., 0., 0., 0., 0., 0., 0., 0., 0.],
        [0., 0., 0., 0., 0., 0., 0., 0., 0., 0.],
        [0., 0., 0., 0., 0., 0., 0., 0., 0., 0.],
        [0., 0., 0., 0., 0., 0., 0., 0., 0., 0.],
        [0., 0., 0., 0., 0., 0., 0., 0., 0., 0.],
        [0., 0., 0., 0., 0., 0., 0., 0., 0., 0.]])
>>>
```

Memory mapping: NumPy

```
>>> import numpy as np
>>> mm = np.memmap('tmp.raw', mode='w+', shape=(10, 10), dtype='float64')
>>> mm.dtype
dtype('float64')
>>> mm.shape
(10, 10)
```

Memory mapping: NumPy

```
>>> import numpy as np
>>> mm = np.memmap('tmp.raw', mode='w+', shape=(10, 10), dtype='float64')
>>> mm[:5, :5] = 1
>>> mm
memmap([[1., 1., 1., 1., 1., 0., 0., 0., 0., 0.],
        [1., 1., 1., 1., 1., 0., 0., 0., 0., 0.],
        [1., 1., 1., 1., 1., 0., 0., 0., 0., 0.],
        [1., 1., 1., 1., 1., 0., 0., 0., 0., 0.],
        [1., 1., 1., 1., 1., 0., 0., 0., 0., 0.],
        [0., 0., 0., 0., 0., 0., 0., 0., 0., 0.],
        [0., 0., 0., 0., 0., 0., 0., 0., 0., 0.],
        [0., 0., 0., 0., 0., 0., 0., 0., 0., 0.],
        [0., 0., 0., 0., 0., 0., 0., 0., 0., 0.],
        [0., 0., 0., 0., 0., 0., 0., 0., 0., 0.]])
>>>
```

Note: this changes the saved file too!

Memory mapping: NumPy

```
>>> import numpy as np
>>> mm = np.memmap('tmp.raw', mode='r')
```

File name

Mode

‘r’ = read only

Memory mapping: NumPy

```
>>> import numpy as np
>>> mm = np.memmap('tmp.raw', mode='r')
>>> mm[0] = 2
Traceback (most recent call last):
  File "<stdin>", line 1, in <module>
ValueError: assignment destination is read-only
```

Memory mapping: NumPy

```
>>> import numpy as np
>>> mm = np.memmap('tmp.raw', mode='r+')
```

File name

Mode

‘r+’ = read and write

Memory mapping: NumPy

```
>>> import numpy as np
>>> mm = np.memmap('tmp.raw', mode='r+')
>>> mm[0] = 2
>>>
```

Memory mapping: NumPy

```
>>> import numpy as np
>>> mm = np.memmap('tmp.raw', mode='r+')
>>> mm[0] = 2
>>> mm.shape
(800,)
>>>
```

???

Memory mapping: NumPy

```
>>> import numpy as np
>>> mm = np.memmap('tmp.raw', mode='r+')
>>> mm[0] = 2
>>> mm.shape
(800,)
>>> mm.dtype
dtype('uint8')
>>> mm[:26]
memmap([  2,   0,   0,   0,   0,   0, 240,  63,   0,   0,   0,   0,   0,
          0, 240,  63,   0,   0,   0,   0,   0,   0, 240,  63,   0,   0],
       dtype=uint8)
```

Except the 2!

Garbage data

Default dtype is uint8 – we are reading raw data!

float64 = 8 bytes

10 * 10 * 8 = 800

Memory mapping: NumPy

```
>>> import numpy as np
>>> mm = np.memmap('tmp.raw', mode='r+')
>>> mm[0] = 2
>>> mm.shape
(800,)
>>> mm.d
dtype('u
>>> mm[:
memmap([
          0, 240,  63,   0,   0,   0,   0,   0,   0, 240,
       dtype=uint8)
```

Default dtype is uint8 – we are reading raw data!

float64 = 8 bytes

Moral: must specify shape and data type for memmaps

Garbage data

Memory mapping: NumPy

```
>>> import numpy as np
>>> mm = np.memmap('tmp.raw', mode='r+', dtype='float64', shape=(10, 10))
>>> mm
memmap([[1., 1., 1., 1., 1., 0., 0., 0., 0., 0.],
        [1., 1., 1., 1., 1., 0., 0., 0., 0., 0.],
        [1., 1., 1., 1., 1., 0., 0., 0., 0., 0.],
        [1., 1., 1., 1., 1., 0., 0., 0., 0., 0.],
        [1., 1., 1., 1., 1., 0., 0., 0., 0., 0.],
        [0., 0., 0., 0., 0., 0., 0., 0., 0., 0.],
        [0., 0., 0., 0., 0., 0., 0., 0., 0., 0.],
        [0., 0., 0., 0., 0., 0., 0., 0., 0., 0.],
        [0., 0., 0., 0., 0., 0., 0., 0., 0., 0.],
        [0., 0., 0., 0., 0., 0., 0., 0., 0., 0.]])
```

Can now read and write like normal NumPy array

Note: changes are written out to disk! Be careful!

Quiz time!

New and shiny

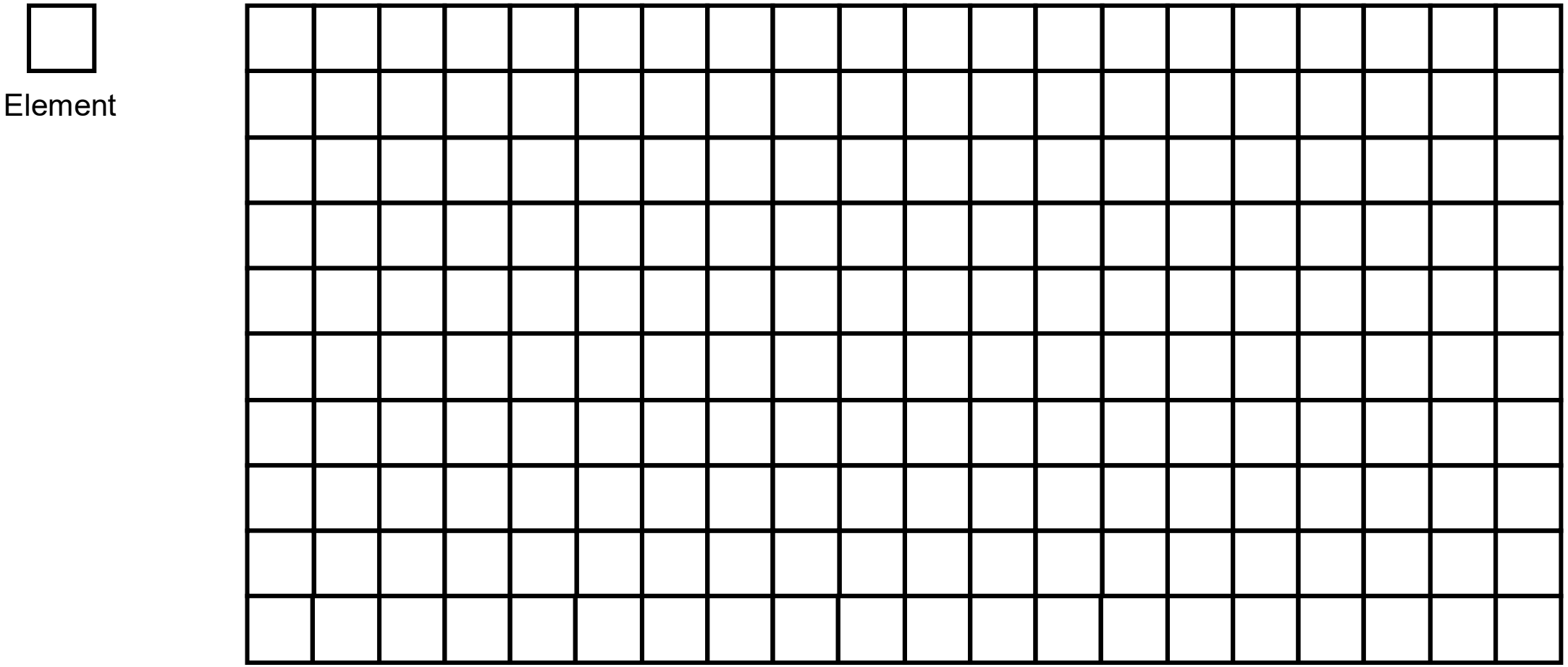
Zarr

HDF5

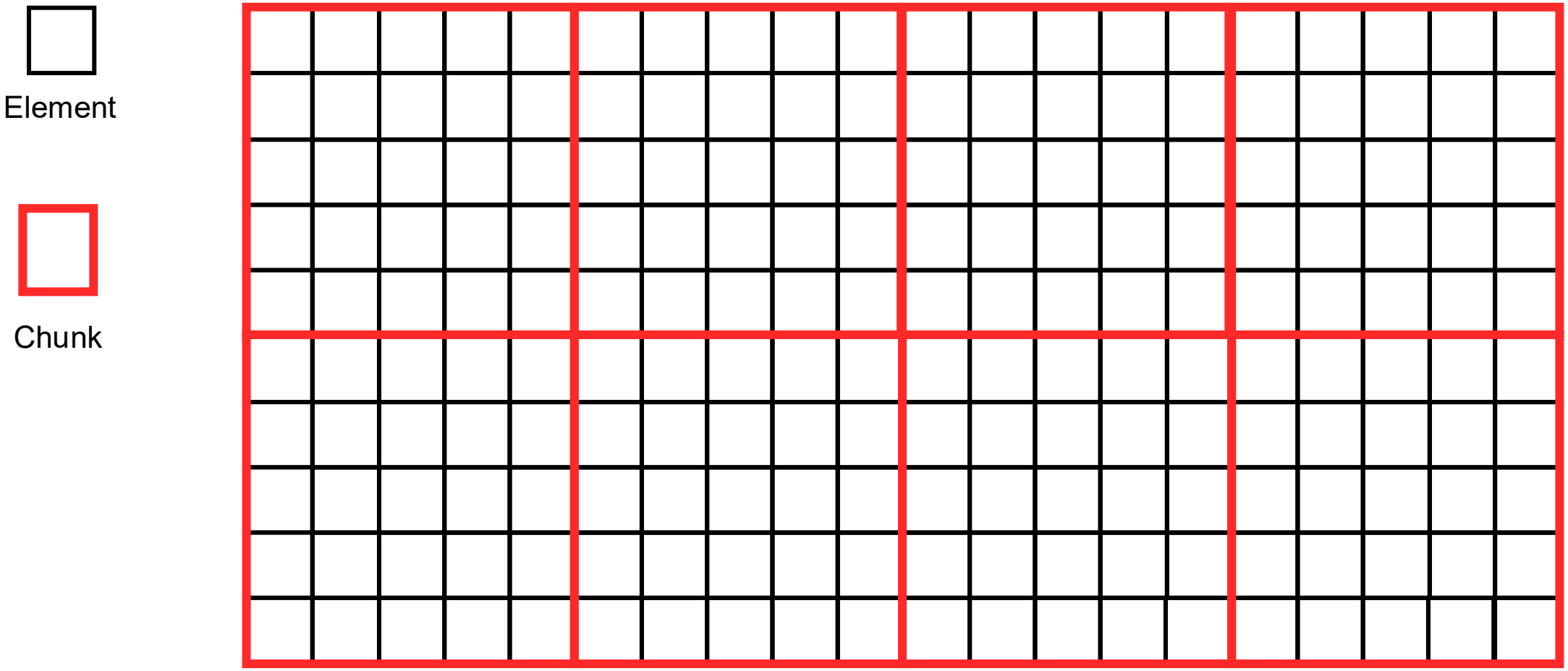
Old and established

Zarr is a file storage format for chunked, compressed, N-dimensional arrays based on an open-source specification.

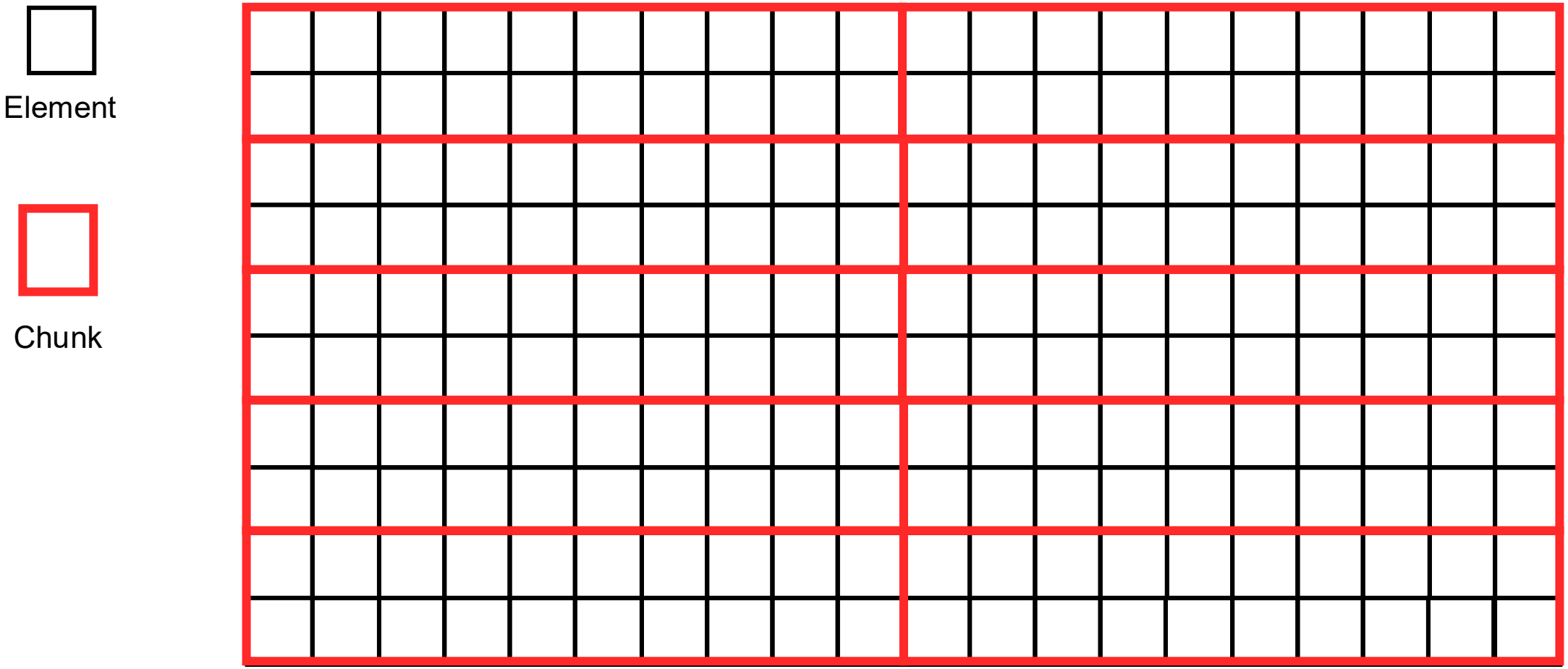
Chunked, compressed, N-dimensional arrays



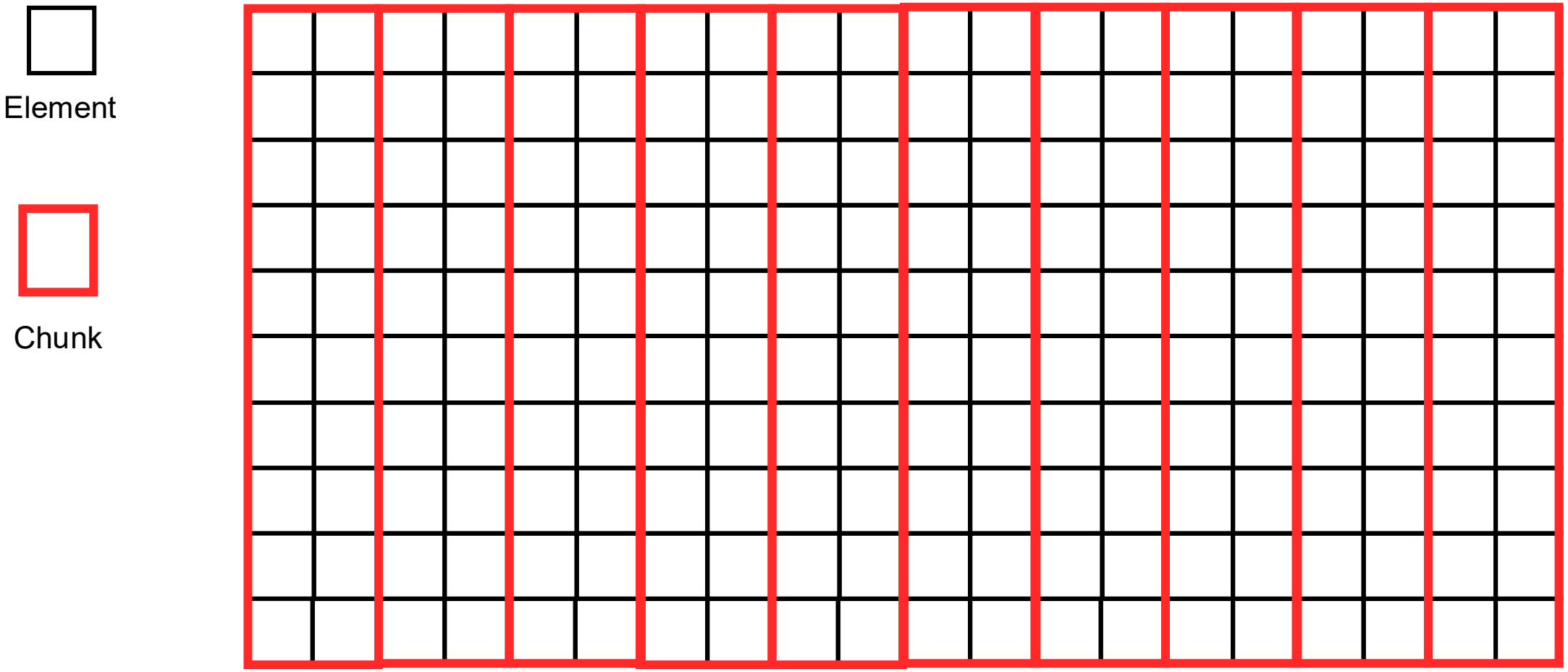
Chunked, compressed, N-dimensional arrays



Chunked, compressed, N-dimensional arrays



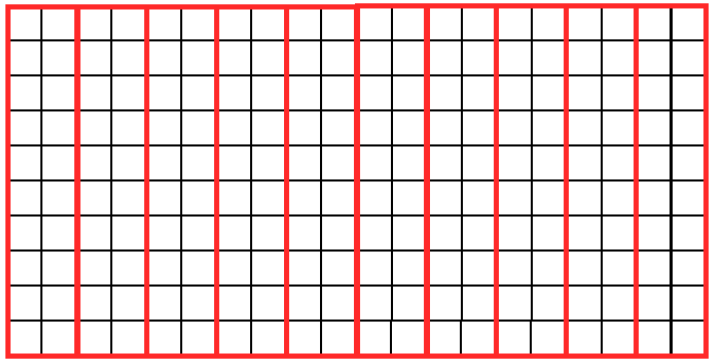
Chunked, compressed, N-dimensional arrays



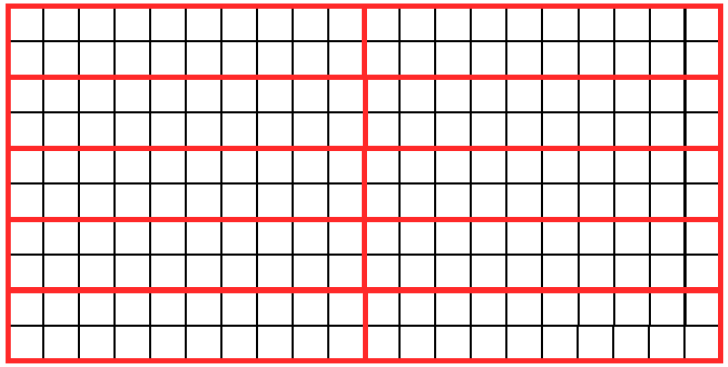
Chunked, compressed, N-dimensional arrays

Chunking allows us to tune the storage pattern to our access pattern

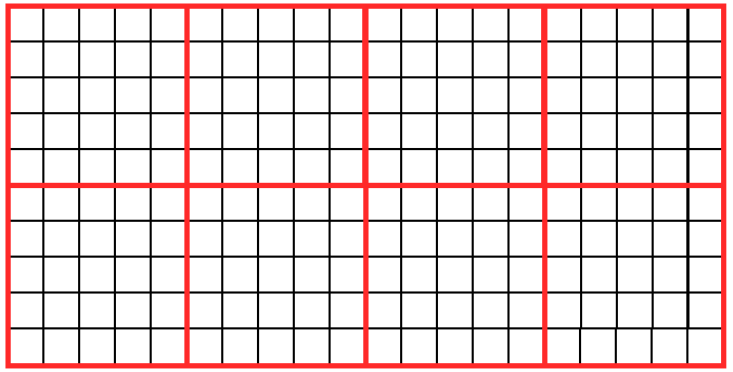
Good for “column-wise”



Good for “row-wise”



Good for “neighborhood-wise”



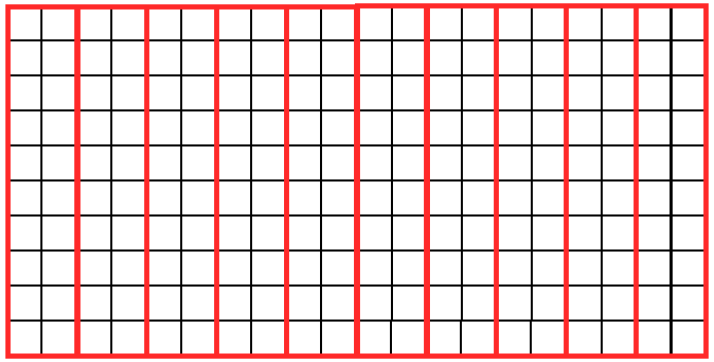
<u>Chunked</u>, compressed, N-dimensional arrays

Chunking ***<u>forces</u>*** us to tune the storage pattern to our access pattern

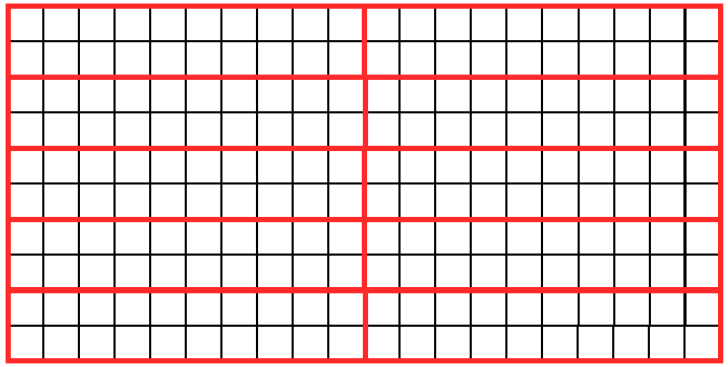
Good performance depends on good chunk size and shape

Should be at least 1 MB uncompressed

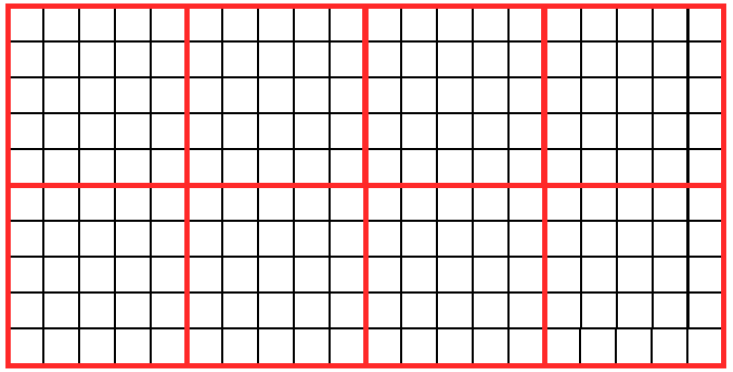
Good for “column-wise”



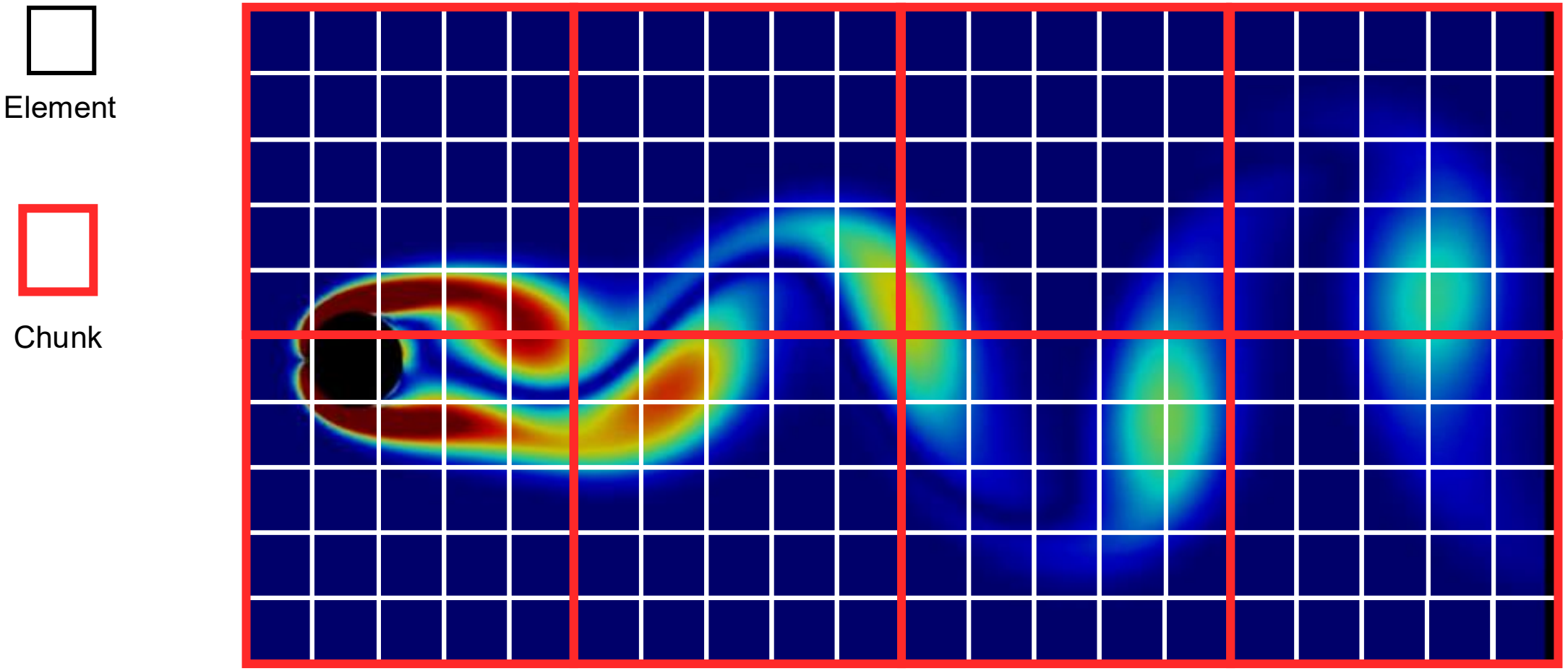
Good for “row-wise”



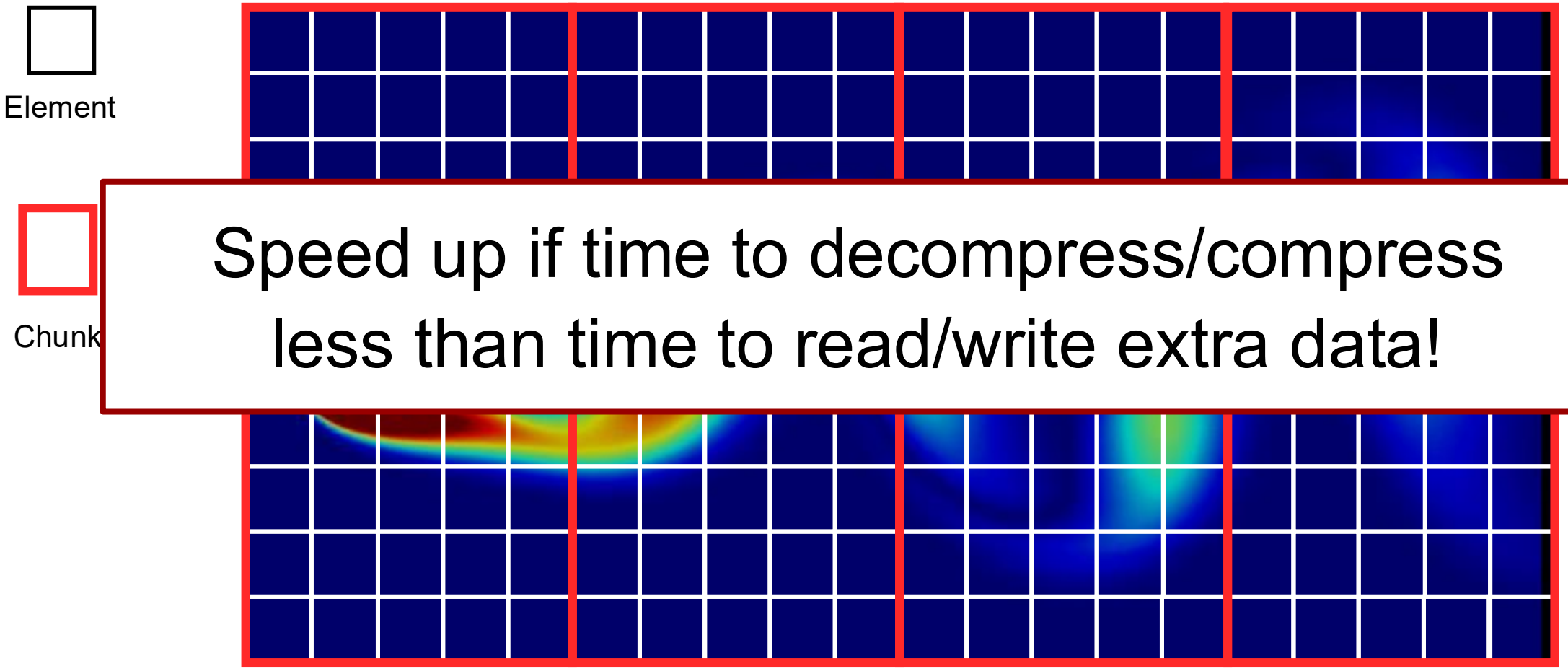
Good for “neighborhood-wise”



Chunked, compressed, N-dimensional arrays

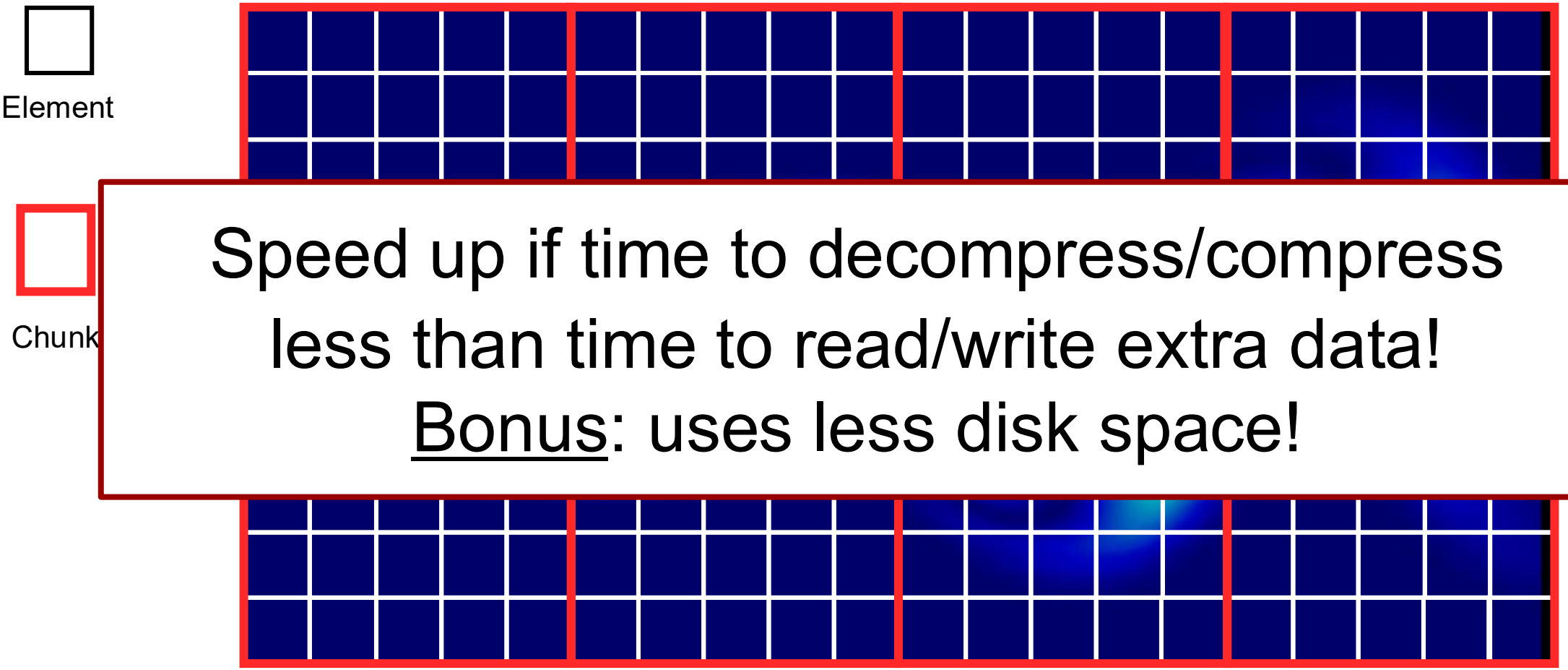


Chunked, compressed, N-dimensional arrays



Speed up if time to decompress/compress less than time to read/write extra data!

Chunked, compressed, N-dimensional arrays



Speed up if time to decompress/compress less than time to read/write extra data!

Bonus: uses less disk space!

Zarr

```
>>> import zarr
>>> array = zarr.open(‘mandelbrot.zarr’)
```

Zarr

```
>>> import zarr
>>> array = zarr.open(‘mandelbrot.zarr’)
>>> array.info
Type               : zarr.core.Array
Data type          : int32
Shape              : (1000, 1000)
Chunk shape        : (200, 200)
Order              : C
Read-only          : False
Compressor         : Blosc(cname='lz4', clevel=5, shuffle=SHUFFLE, blocksize=0)
Store type         : zarr.storage.DirectoryStore
No. bytes          : 4000000 (3.8M)
No. bytes stored   : 95666 (93.4K)
Storage ratio      : 41.8
Chunks initialized : 25/25
```

Zarr

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Shape of whole array

Zarr

```
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Chunks initialized : 25/25
```

Shape of each chunk

Zarr

```
>>> import zarr
>>> array = zarr.open(‘mandelbrot.zarr’)
>>> array.info
Type                  : zarr.core.Array
Data type             : int32
Shape                 : (1000, 1000)
Chunk shape           : (200, 200)
Order                 : C
Read-only             : False
Compressor            : Blosc(cname='lz4', clevel=5, shuffle=SHUFFLE, blocksize=0)
Store type            : zarr.storage.DirectoryStore
No. bytes             : 4000000 (3.8M)
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Chunks initialized    : 25/25
```

Data order for each chunk.
C = row-wise, F = Fortran = column-wise

Zarr

```
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>>> array = zarr.open(‘mandelbrot.zarr’)
>>> array.info
Type               : zarr.core.Array
Data type          : int32
Shape              : (1000, 1000)
Chunk shape        : (200, 200)
Order              : C
Read-only          : False
Compressor         : Blosc(cname='lz4', clevel=5, shuffle=SHUFFLE, blocksize=0)
Store type         : zarr.storage.DirectoryStore
No. bytes          : 4000000 (3.8M)
No. bytes stored   : 95666 (93.4K)
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Chunks initialized : 25/25
```

Chunk compression algorithm

Zarr

```
>>> import zarr
>>> array = zarr.open(‘mandelbrot.zarr’)
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Data type          : int32
Shape              : (1000, 1000)
Chunk shape        : (200, 200)
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Read-only          : False
Compressor         : Blosc(cname='lz4', clevel=5, shuffle=SHUFFLE, blocksize=0)
Store type         : zarr.storage.DirectoryStore
No. bytes          : 4000000 (3.8M)
No. bytes stored   : 95666 (93.4K)
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Chunks initialized : 25/25
```

Storage method

Zarr

…but not HDF5

```
>>> import zarr
>>> array = zarr.open('mandelbrot.zarr')
>
T
D
S
C
O
R
C
S
N
N
Storage ratio      : 41.8
Chunks initialized : 25/25
```

Zarr supports several storage backends, e.g.:

- DirectoryStore (standard for persistent storage)
- MemoryStore (everything in memory)
- SQLiteStore (backed by SQLite database)
- MongoDBStore (backed by MongoDB)
- Amazon S3, Google Cloud Storage, Azure Blob Storage
- LRUStoreCache (cache results from slower store)

Zarr

```
>>> import zarr
>>> array = zarr.open(‘mandelbrot.zarr’)
>>> array.info
Type               : zarr.core.Array
Data type          : int32
Shape              : (1000, 1000)
Chunk shape        : (200, 200)
Order              : C
Read-only          : False
Compressor         : Blosc(cname='lz4', clevel=5, shuffle=SHUFFLE, blocksize=0)
Store type         : zarr.storage.DirectoryStore
No. bytes          : 4000000 (3.8M)     Bytes needed to store full array
No. bytes stored   : 95666 (93.4K)      Bytes actually used after compression
Storage ratio      : 41.8
Chunks initialized : 25/25
```

Zarr

```
>>> import zarr
>>> array = zarr.open(‘mandelbrot.zarr’)
>>> array.info
Type               : zarr.core.Array
Data type          : int32
Shape              : (1000, 1000)
Chunk shape        : (200, 200)
Order              : C
Read-only          : False
Compressor         : Blosc(cname='lz4', clevel=5, shuffle=SHUFFLE, blocksize=0)
Store type         : zarr.storage.DirectoryStore
No. bytes          : 4000000 (3.8M)    Bytes needed to store full array
No. bytes stored   : 95666 (93.4K)     Bytes actually used after compression
Storage ratio      : 41.8              (No. bytes) / (No. bytes stored)
Chunks initialized : 25/25
```

Zarr

```
>>> import zarr
>>> array = zarr.open(‘mandelbrot.zarr’)
>>> array.info
Type               : zarr.core.Array
Data type          : int32
Shape              : (1000, 1000)
Chunk shape        : (200, 200)
Order              : C
Read-only          : False
Compressor         : Blosc(cname='lz4', clevel=5, shuffle=SHUFFLE, blocksize=0)
Store type         : zarr.storage.DirectoryStore
No. bytes          : 4000000 (3.8M)
No. bytes stored   : 95666 (93.4K)
Storage ratio      : 41.8
Chunks initialized : 25/25
```

Zarr

```
>>> import zarr
>>> empty = zarr.open('empty.zarr', mode='w', shape=(100,100), chunks=(20, 20))
```

File name

Mode

‘w’ = create file

Array shape

Chunk shape

Zarr

```
>>> import zarr
>>> empty = zarr.open('empty.zarr', mode='w', shape=(100,100), chunks=(20, 20))
>>> empty.info
Type               : zarr.core.Array
Data type          : float64
Shape              : (100, 100)
Chunk shape        : (20, 20)
Order              : C
Read-only          : False
Compressor         : Blosc(cname='lz4', clevel=5, shuffle=SHUFFLE, blocksize=0)
Store type         : zarr.storage.DirectoryStore
No. bytes          : 80000 (78.1K)
No. bytes stored   : 339
Storage ratio      : 236.0
Chunks initialized : 0/25
```

Great for sparse data!

Only some metadata is actually stored!

No chunks initialized

Zarr: groups

```
>>> import zarr
>>> data = zarr.open(‘data.zarr’, mode=‘w’)
>>> data.create_group(‘group1’)
>>> data.create_group(‘group2’)
>>> data[‘group1’].create_group(‘arrays’)  # Nested group
>>> data.tree()
/
 ├── group1
 │    └── arrays
 └── group2
```

Groups act like folders in a file system
(also available in HDF5)

Zarr: create array in group

```
>>> import zarr
>>> data = zarr.open(‘data.zarr’, mode=‘w’)
>>> data.create_group(‘group1’)
>>> data.create_group(‘group2’)
>>> data[‘group1’].create_group(‘arrays’)  # Nested group
>>> data[‘group1/arrays’].zeros(‘array1’, shape=(100, 100), chunks=(50, 20))
```

Array “path” | Array name | Array shape | Chunk shape

Zarr: assign array to group

```
>>> import zarr
>>> data = zarr.open(‘data.zarr’, mode=‘w’)
>>> data.create_group(‘group1’)
>>> data.create_group(‘group2’)
>>> data[‘group1’].create_group(‘arrays’)  # Nested group
>>> data[‘group1/arrays’].zeros(‘array1’, shape=(100, 100), chunks=(50, 20))
>>> data['group1/arrays/array2'] = zarr.zeros(shape=(100, 100), chunks=(50, 20))
```

Array “path” Array name Array shape Chunk shape

Zarr: list contents

```
>>> import zarr
>>> data = zarr.open(‘data.zarr’, mode=‘w’)
>>> data.create_group(‘group1’)
>>> data.create_group(‘group2’)
>>> data[‘group1’].create_group(‘arrays’)  # Nested group
>>> data[‘group1/arrays’].zeros(‘array1’, shape=(100, 100), chunks=(50, 20))
>>> data['group1/arrays/array2'] = zarr.zeros(shape=(100, 100), chunks=(50, 20))
>>> data.tree()
/
 ├── group1
 │    └── arrays
 │        ├── array1 (100, 100) float64
 │        └── array2 (100, 100) float64
 └── group2
```

Zarr works as dataset storage, not just arrays

Zarr: internal format

```
$ ls
test.zarr
$ ls test.zarr
group1  group2
```

Zarr: internal format

```
$ ls
test.zarr
$ tree test.zarr
test.zarr/
├── group1
│   └── arrays
│       ├── array1
│       └── array2
│           ├── 0.0
│           ├── 0.1
...
│           ├── 1.2
│           ├── 1.3
│           └── 1.4
└── group2
```

Chunks

Zarr (with DirectoryStore) stores data in an *actual* directory (/folder)

Groups and arrays are sub-folders

Chunks are separate files

Zarr: internal format

```
$ ls
test.zarr
$ ls -a test.zarr
.  ..  group1  group2  .zgroup
$ ls -a test.zarr/group1/arrays/array1
.  ..  .zarray
```

Zarr (with DirectoryStore) stores data in an *actual* directory (/folder)

Groups and arrays are sub-folders

Chunks are separate files

Meta data are in files

Zarr: internal format

```
$ ls
test.zarr
$ ls -a test.zarr
.  ..  group1  group2  .zgroup
$ ls -a test.zarr/group1/arrays/array1
.  ..  .zarray
```

Consequence:
Can read *and* write to chunks in parallel!
(write not possible in HDF5)

Zarr (with DirectoryStore) stores data in an *actual* directory (/folder)

Groups and arrays are sub-folders

Chunks are separate files

Meta data are in files

Zarr: indexing

```
>>> import zarr
>>> array = zarr.open(‘mandelbrot.zarr’)
>>> array[:5, :10]
array([[0, 0, 0, 0, 0, 0, 0, 0, 0, 0],
       [0, 0, 0, 0, 0, 0, 0, 0, 0, 0],
       [0, 0, 0, 0, 0, 0, 0, 0, 0, 0],
       [0, 0, 0, 0, 0, 0, 0, 0, 0, 0],
       [0, 0, 0, 0, 0, 0, 0, 0, 0, 0]], dty
>>> type(array[:5, :10])
<class 'numpy.ndarray'>
```

Can index like NumPy array and get a NumPy array(!) out.

- Python indexing: [start:end:step]
- Boolean indexing: [True, True, …]
- Array of indexes: [1, 2, 3]

Note: be careful for large arrays!

Zarr: block indexing

```
>>> import zarr
>>> array = zarr.open(‘mandelbrot.zarr’)
>>> array.shape
(1000, 1000)
>>> array.chunks
(200, 200)
>>> array.cdata_shape
(5, 5)
```

Shape of whole array

Shape of each chunk

Chunks per dimension

Zarr: block indexing

```
>>> import zarr
>>> array = zarr.open(‘mandelbrot.zarr’)
>>> array.blocks[0, 1]
array([[0, 0, 0, ..., 0, 0, 0],
       [0, 0, 0, ..., 0, 0, 0],
       [0, 0, 0, ..., 0, 0, 0],
       ...,
       [1, 1, 1, ..., 2, 2, 2],
       [1, 1, 1, ..., 2, 2, 2],
       [1, 1, 1, ..., 2, 2, 2]], dtype=int32)
>>> array.blocks[0, 1].shape
(200, 200)
>>> array.chunks
(200, 200)
```

Zarr: block indexing

```
>>> import zarr
>>> array = zarr.open(‘mandelbrot.zarr’)
>>> array.blocks[0, 1]
array([[0, 0, 0, ..., 0, 0, 0],
       [0, 0, 0, ..., 0, 0, 0],
       [0, 0, 0, ..., 0, 0, 0],
       ...,
       [1, 1, 1, ..., 2, 2, 2],
       [1, 1, 1, ..., 2, 2, 2],
       [1, 1, 1, ..., 2, 2, 2]], dtype=int32)
>>> (array.blocks[0, 1] == array[:200, 200:400]).all()
True
```

Zarr: block indexing

```
>>> import zarr
>>> array = zarr.open(‘mandelbrot.zarr’)
>>> array.blocks[0, 1] = 8   # Could also assign array instead of single number
>>> array.blocks[0, 1]
array([[8, 8, 8, ..., 8, 8, 8],
       [8, 8, 8, ..., 8, 8, 8],
       [8, 8, 8, ..., 8, 8, 8],
       ...,
       [8, 8, 8, ..., 8, 8, 8],
       [8, 8, 8, ..., 8, 8, 8],
       [8, 8, 8, ..., 8, 8, 8]], dtype=int32)
```

Quiz time!

Today’s exercise

Computing in memory constrained environments

- Pandas with chunked reading
- Mandelbrot with NumPy memory maps
- Mandelbrot with Zarr arrays

Useful concepts

Pandas

```
df = pd.read_csv('data.csv', chunksize=C)
```

NumPy memmap

```
mode = 'w+'        # Make new file
mode = 'r+'        # Read and write
mode = 'r'         # Read only
mm = np.memmap('file.raw', mode=mode,
                   shape=(N,M), dtype='int32')
```

Zarr

```
array = zarr.open('mandelbrot.zarr')
array.shape              # Array shape
array.chunks             # Chunk shape
array.cdata_shape        # Chunks per dimension
array.blocks[0, 1]       # Chunk access
```

Change to work node

linuxsh

Submit job script

bsub < submit.sh

Job status

bstat / bjobs

Check job output

bpeek / bpeek <JOBID>

Kill job

bkill <JOBID>